

A COMPARATIVE ANALYSIS OF OTTOMAN-TURKISH MAKAMS AND HINDUSTANI RAGAS

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ABSTRACT

This paper presents a computational comparison of two Eastern music traditions: North Indian (Hindustani) Ragas and Ottoman-Turkish Makams, using symbolic MusicXML datasets and the MelodyShape similarity framework. Both traditions share a common approach to melody through large systems of different modes, each with its own identity and rules. By applying the ShapeH and Time melodic similarity metrics, melodic structure was examined at multiple levels, from full composition comparisons down to individual melodic fragment pairs. The analysis focused on three Hindustani Ragas, compared against a corpus of Turkish Makam compositions. Results showed that Raga Bhairav consistently produced strong matches with several Turkish Makams, a finding supported by shared structural properties at the intervallic and tetrachordal level. These findings provided computational evidence for theoretical claims about structural similarities between Eastern modal traditions, while also highlighting the limitations of symbolic data in capturing the full richness of these musical systems. Open-source code is available at <https://github.com/Sangarshanan/makam-and-ragas>

1. INTRODUCTION

Eastern musical traditions have long been a subject of scholarly interest, and in recent years they have gained increasing attention in Music Information Retrieval (MIR) research. Among these traditions, Ottoman-Turkish Makams and North Indian (Hindustani) Ragas stand out for their richness and melodic approach through many modes: both systems use a wide variety of intervals, distinct melodic patterns, and complex rhythmic structures.

These two traditions are not historically unrelated. From the 11th century A.D., Arabic, Persian and Turkic cultures began to exert influence on North India, contributing to the development of new forms in Indian literature, art, and music. Despite their geographical distance, both traditions share a common function: organizing melody

through systems of many distinct modes, each with its own identity and rules.

The growing availability of symbolic datasets for both traditions has made it possible to study these properties computationally. MIR research has begun exploring melodic patterns in these traditions with digital tools, and the publication of new datasets has opened further opportunities for comparison and analysis. This paper examines the two traditions side by side, to explore their varieties and possible correlations and to demonstrate the value of symbolic data in studying these traditions computationally.

To do this, the melodic behavior of Makam and Raga was compared in musical pieces through the approach of pitch curve analysis, which captures the shape and direction of melodic movement and reveals the intervallic preferences of each tradition.

2. RELATED WORK

The study of the relationship between Arabic and Indian cultures has attracted scholarly interest given their historical connections.

[1] explored the similarity of North Indian Hindustani Raga Zeelaf created by Amīr Khusrow with Arabian traditional makams with the goal of proving the exchange of musical knowledge between the Persian, Arabic and Indian cultures by doing signal analysis. Their method involved recording audio performances of Raga Zeelaf in its two forms (Bhairav and Asavari Thaat), extracting pitch information through F0 estimation and steady-state detection, and mapping the resulting frequencies to standard note notations. The same process was applied to ten widely performed Arabic Maqams. By comparing the transcribed note sets of both traditions, it was found that the Bhairav form of Zeelaf matches with Maqams Shadd 'Araban and Hijaz Kar, while the Asavari form partially matches with Nahawand, Nahawand Kurd and Rast.

[2] provided a basis for comparing Ottoman-Turkish Makam and North Indian Raga as two modally rich traditions in Eastern music. Through a multi-level structural analysis, he identified seventeen mechanisms responsible for generating modal variety in both systems, ranging from intervallic content and degree hierarchy to obligatory melodic phrases and modulation practices. The author's main idea was that neither system can be characterized by its scale alone. The modal identity in both traditions is



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created from a combination of behavioral, structural, and phraseological features that go far beyond pitch content.

[3] evaluated the performance of various symbolic similarity measures in identifying melodic occurrences, such as phrases and motifs, within a large corpus of Dutch folk songs. Their study compared eight distinct computational approaches: traditional distance-based metrics (correlation, city-block, and Euclidean distance), alignment methods (local, B-spline and Implication-Realization structure alignment), and pattern discovery techniques (wavelet transforms and the geometric Structure Induction algorithm, SIAM). By testing these measures against expert human annotations of phrase similarity, the authors found that simple distance metrics, often used in earlier ethnomusicological research, performed poorly in identifying melodic variants. In contrast, local alignment and the geometric SIAM algorithm demonstrated the highest efficiency, performing at a level nearly comparable to human annotators.

3. METHODOLOGY

To explore and visualize the similarities between the two traditional genres, pitch curves were extracted and used to examine tonal variety and correlations between melodic contours.

3.1 Datasets

For both traditions, XML format datasets were used to explore how symbolic representation can be applied to this type of analysis.

For Ragas, the SANGEET dataset [4] was used, which consists of 116 XML representations of compositions belonging to three Ragas: Bhairav (42), Todi (39), and Poorvi (35). The compositions were taken from the pioneer Pt. Vishnu Narayan Bhatkhande’s book *Hindustani Sangeet Paddhati*, Kramik Pustak Malika.

For Turkish Makams, the SymbTr dataset [5] was used, which contains 1,700 musical pieces of two Makam styles: classical Turkish music and Turkish folk music. SymbTr features 155 Makams, 100 Usuls and 48 forms. The main source of data in SymbTr was TRT and other trustworthy archives, where almost all pieces were entered using the AEU notation.

3.2 Preprocess

To prepare the datasets for the experiment two different parsers were created, since their XML format is different.

The Turkish Makam corpus uses the standard MusicXML partwise format, with each file having a structured filename form of makam–form–usul–title–composer.xml. This allowed filtering by Makam or Usul directly from the filename. Notes were then parsed measure by measure, with rests and tied continuation notes skipped to avoid double-counting. Each note’s pitch was converted into an absolute semitone value by combining its letter name, octave, and alteration value. Microtonal adjustments were kept as decimal numbers so that the microtonal tuning characteristics of Makam music are not lost. The tonic of

each piece, known as the Karar, was identified as the last pitched note of the composition, following standard practice in the corpus.

The Hindustani Raga dataset uses a custom Swarlipi XML format based on Bhatkhande notation [6], where notes are encoded as single ASCII characters rather than standard pitch names. Each character was mapped to its corresponding note name through a fixed lookup table covering all twelve scale degrees, including their komal (flat) and shuddha (natural) variants. For example, lowercase r represents shuddha Re while uppercase R represents komal Re. Before note counting, the raw sheet content was cleaned by stripping formatting symbols, markup tags, and ornament delimiters such as meend and murki markers, which would otherwise interfere with note identification. Once cleaned, each note character was identified by converting it to its ASCII code point and matching it against the known note codes. This produced a note frequency count for each composition, equivalent to a pitch class histogram, which served as the basis for further melodic analysis alongside the Raga label extracted from the XML metadata.

3.3 Methodology

For the melodic similarity computation, the MelodyShape method [7] was used. MelodyShape is a symbolic melodic similarity algorithm that compares melodies by fitting polynomial curves to pitch and timing contours, then aligning them via sequence alignment.

The core idea is that a melody is chunked into overlapping spans of n notes. Each span is represented by a polynomial curve fitted over its pitch contour and timing ratios. Two melodies are then compared by aligning their span sequences using dynamic programming. Three invariances follow from this:

- **Transposition invariance:** pitches are expressed as intervals relative to the span’s first note; the derivative of a transposed melody is identical to the original
- **Time-scale invariance:** durations become ratios within a span, so a melody at double tempo maps to the same curve
- **Position invariance:** the hybrid alignment finds the best-matching subsequence, so a query fragment can match anywhere in a longer piece

ShapeH and Time similarity metrics were then used to compare melodic curves across the two corpora.

3.3.1 ShapeH

ShapeH compares melodies on melodic contour, ignoring absolute pitch and timing. Each melody is segmented into overlapping 3-note spans, fitted with second-degree polynomials that are then differentiated to capture the direction and rate of melodic movement. Two spans are considered similar if their derivative signs match at both ends.

ShapeH incorporates an IDF-inspired weighting scheme where $f(n)$ denotes the frequency of span n in the collection, so that sharing a rare melodic pattern contributes more to the similarity score than sharing a common one.

The final score is computed via dynamic programming alignment, where $H(i, j)$ denotes the best alignment score between the first i spans of melody A and the first j spans of B :

$$H(i, j) = \max \left\{ \begin{array}{l} H(i-1, j-1) + s(a_i, b_j) \\ H(i-1, j) + s(a_i, -) \\ H(i, j-1) + s(-, b_j) \end{array} \right\} \quad (1)$$

$$\text{ShapeH}(A, B) = \max_{i,j} H(i, j) \quad (2)$$

This is particularly suited to Makam and Raga music, where opening phrases are primary markers of modal identity [2].

3.3.2 Time

The Time similarity system [7] extends ShapeH by incorporating rhythm. It uses 4-note spans fitted with degree-3 polynomials (differentiated to degree 2), capturing more complex melodic shapes that can change direction twice within a span. A second polynomial is fitted to the note durations within each span, normalized to a fixed length of 1, making the system both time-scale and transposition invariant.

The same hybrid alignment from ShapeH is then applied. This makes Time particularly suited to Makam and Raga music, where rhythmic cycles (*usul* and *tala*) are as structurally significant as melodic contour.

3.4 Analysis

For the analysis, all files were characterized and compared, organized by their Raga and Makam. For each melody, overlapping spans of consecutive notes were extracted and used as the basic unit of comparison. For every possible Raga-Makam file combination, a pairwise melodic similarity score was computed.

Once all pairwise scores were computed, they were aggregated by Raga and Makam name across all file combinations, and summary statistics were computed for each Raga-Makam pair. Each raw score was mapped to $[0, 1]$ using (3). Finally, the best-matching Makam for each Raga was identified by aggregating similarity scores.

$$\text{similarity} = \frac{1}{1 + \max(0, -\text{score})} \quad (3)$$

The five most similar composition pairs were chosen for visualization. Each selected span was converted into a normalized pitch curve by evaluating its fitted polynomial at evenly spaced time points and scaling both axes to $[0, 1]$, allowing spans of different lengths and pitch ranges to be compared purely on the basis of their melodic shape. The Raga curve was plotted in blue and the Makam curve in

red, with the area between them shaded in gray to highlight their differences. Geometric distances MAD (Mean Absolute Distance) and RMSE (Root Mean Square Error) were also computed as additional distance measures.

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (5)$$

After ranking the melodies by ShapeH, the top-ranked results were re-ranked by Time score following the ShapeTime system suggested in [7].

The ShapeTime ranking was computed by applying the two-stage algorithm to each Raga file against the full Makam collection. First, ShapeH scores were computed for all Raga-Makam pairs and the Makams were sorted by melodic contour similarity. Then, Time scores were computed only for the shortlisted Makams against the Raga with the highest similarity score, making the pipeline more efficient by avoiding the more expensive time-dimension computation for dissimilar pairs.

The shortlisted Makams were then re-ranked by their Time score. Each Makam received a final ShapeTime rank, and a ShapeTime score was assigned as the Time score for re-ranked candidates and the ShapeH score for the rest. The results were aggregated across all Raga files belonging to the same modal class, producing summary statistics for each Raga-Makam pair, allowing the most consistently similar Makams to be identified across multiple compositions of the same Raga.

4. RESULTS

The top-5 highest-scoring span pairs were retained and plotted in Figure 2, showing each Raga-Makam fragment pair as normalized pitch curves in the $[0, 1] \times [0, 1]$ plane, with gray shading highlighting geometric differences and ShapeH, MAD and RMSE reported per pair. This revealed the specific melodic gestures shared between the two traditions beyond the file-level score.

Most top results were detected between Raga Bhairav and Hüseyni Türkü Makam, as shown in Figure 1.

ShapeTime was then applied to all Raga Bhairav compositions to re-rank Makams by incorporating the time metric. Top results are displayed in Table 1.

The strong melodic similarity between Raga Bhairav and several Turkish Makams can be explained by shared structural characteristics at both the intervallic and tetrachordal level. The most fundamental point of convergence is the role of the flattened second degree. In Bhairav, the komal Re (flat 2nd) and komal Dha (flat 6th) are not merely scale tones but the melodic centers of the Raga. The half-step movement from komal Re down to Sa creates a characteristic tension and pull toward the tonic that defines the Raag's identity [8]. This same interval logic is central to Turkish Makam thinking, where the minor second below

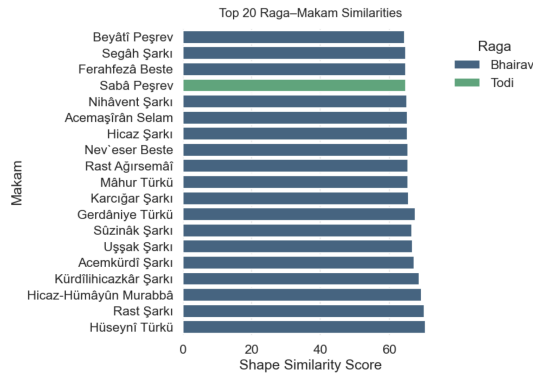


Figure 1. Top 20 Raga-Makam similarity pairs, after computing ShapeH similarity score.

a resting tone is one of the primary tools for generating melodic tension and resolution.

A second structural similarity lies in the tetrachordal organization of the octave. Bhairav divides the octave into two asymmetric tetrachords, where the lower tetrachord (Sa, komal Re, Ga, Ma) contains a minor second followed by a major third, a characteristic of Turkish Makam structure, which is fundamentally built around the stacking and combining of tetrachords. These shared structural properties could possibly explain why Bhairav consistently was the stronger match for several Makams in the computational results of this study.

In contrast to Bhairav, Raga Todi presents a more complex interval structure that maps less cleanly onto Turkish Makam grammar. Todi’s scale contains four simultaneously altered degrees: flat 2nd, flat 3rd, sharp 4th, and flat 6th, making it one of the most chromatically dense Ragas in the Hindustani system [8]. The key structural difference is that its lower tetrachord does not correspond to any standard Makam tetrachord type, making it structurally incompatible with the tetrachordal grammar that defines many Makams.

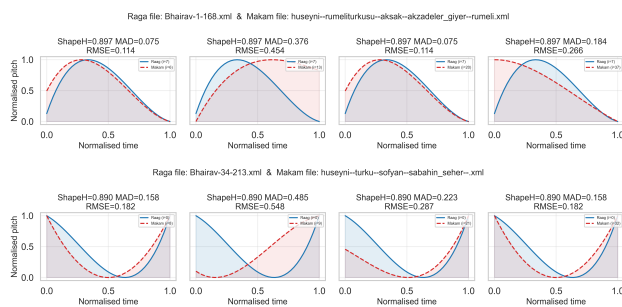


Figure 2. Most similar melodic span pairs between Raag Bhairav and the best-matching Makam. Blue (solid) = Raga, red (dashed) = Makam. Both axes are normalized to $[0, 1]$ for shape-only comparison. Gray shading shows the difference between curves. Titles report ShapeH, MAD and RMSE per pair.

#	Makam	Usul	Avg Rank	ShapeH
1	Sabâ Peşrev	Çenber	34.6	75.98
2	Beyâtî Beste	Sofyan	55.5	74.59
3	Karcıgar Şarkı	Sofyan	56.7	73.26
4	Acem Selâm	Devr-i Kebîr	78.2	72.19
5	Beyâtî Beste	Nîmdevir	88.7	73.67
6	Acemaşîrân	Yürüksemâî	90.4	73.09
7	Nikrîz Fantezi	Yürüksemâî II	95.2	69.77
8	Ferahfezâ Beste	Fireng-i Fer	105.1	71.15
9	Nev’ eser Beste	Zencîr	118.7	72.30
10	Nişâburek Peşrev	Muhammes	125.8	70.68

Table 1. Top 10 Makams most similar to Raag Bhairav, sorted by average ShapeTime rank. A lower Avg Rank indicates the makam consistently ranked as more similar across all Bhairav melody comparisons. ShapeH is the mean melodic shape similarity score (ignoring timing).

5. CONCLUSION

This paper presented a computational melodic comparison between Hindustani Ragas and the Ottoman-Turkish Makam corpus using symbolic MusicXML datasets and the MelodyShape similarity framework. By applying the ShapeH and Time systems, melodic similarity was examined at multiple levels, from full compositions down to individual melodic fragment pairs.

The results showed that Raga Bhairav consistently matched with several Turkish Makams, a finding attributable to shared structural properties at the intervallic and tetrachordal level, in line with the theoretical framework of [2], who argued that Eastern modal traditions share meaningful structural characteristics.

However, this study also highlighted the limitations of symbolic data for this type of analysis. The MusicXML notation could not sufficiently represent the microtonal inflections, pitch bending, and ornamental nuances that are central to modal identity in both traditions. In addition, the Hindustani dataset was limited to only three Ragas, limiting the breadth of cross-traditional conclusions.

Future work could address some of these limitations by combining symbolic analysis with audio-based approaches, allowing the full richness of these traditions to be captured computationally.

6. REFERENCES

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